NUMBER THEORY

NUMBER THEORY

- Counting Numbers ($\mathbb{N}$)

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13,
14, 15, 16, 17, 18, 19, 20, 21, 22, 23,
24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
34, 35, 36, 37, 38, 39, 40, 41, 42, 43,
44, 45, 46, 47, 48, 49, 50, 51, 52, 53,
54, 55, 56, 57, 58, 59, 60, 61, 62, 63,
64, 65, 66, 67, 68, 69, 70, 71, 72, 73,
74, 75, 76, 77, 78, 79, 80, 81, 82, 83,
84, 85, 86, 87, 88, 89, 90, 91, 92, 93,
94, 95, 96, 97, 98, 99, 100, 101, 102,
103, 104, 105, 106, 107, 108, 109,
110, 111, 112, 113, 114, 115, 116,

NUMBER THEORY

- Counting Numbers ($\mathbb{N}$)
- Whole Numbers ($\mathbb{W}$)

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,
13, 14, 15, 16, 17, 18, 19, 20, 21, 22,
23, 24, 25, 26, 27, 28, 29, 30, 31, 32,
33, 34, 35, 36, 37, 38, 39, 40, 41, 42,
43, 44, 45, 46, 47, 48, 49, 50, 51, 52,
53, 54, 55, 56, 57, 58, 59, 60, 61, 62,
63, 64, 65, 66, 67, 68, 69, 70, 71, 72,
73, 74, 75, 76, 77, 78, 79, 80, 81, 82,
83, 84, 85, 86, 87, 88, 89, 90, 91, 92,
93, 94, 95, 96, 97, 98, 99, 100, 101,
102, 103, 104, 105, 106, 107, 108,
109, 110, 111, 112, 113, 114, 115,

NUMBER THEORY

- Counting Numbers ($\mathbb{N}$)
- Whole Numbers ($\mathbb{W}$)
- Integers ($\mathbb{Z}$)

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, …

-1, -2, -3, -4, -5, -6, -7, -8, -9, …

0

NUMBER THEORY

- Counting Numbers ($\mathbb{N}$)
- Whole Numbers ($\mathbb{W}$)
- Integers ($\mathbb{Z}$)
- Rational Numbers ($\mathbb{Q}$)

1/2, 3/4 , 11/7, -6/13,…

NUMBER THEORY

- Counting Numbers ($\mathbb{N}$)
- Whole Numbers ($\mathbb{W}$)
- Integers ($\mathbb{Z}$)
- Rational Numbers ($\mathbb{Q}$)
- Irrational Numbers

e (Euler's Number, 2.71828),

Π (Pi, 3.14159265),

$\sqrt{2}$, $\sqrt[3]{7}$

NUMBER THEORY

- Counting Numbers ($\mathbb{N}$)
- Whole Numbers ($\mathbb{W}$)
- Integers ($\mathbb{Z}$)
- Rational Numbers ($\mathbb{Q}$)
- Irrational Numbers
- Real Numbers ($\mathbb{R}$)

- Rational $\cup$ Irrational

NUMBER THEORY

- Counting Numbers ($\mathbb{N}$)
- Whole Numbers ($\mathbb{W}$)
- Integers ($\mathbb{Z}$)
- Rational Numbers ($\mathbb{Q}$)
- Irrational Numbers
- Real Numbers ($\mathbb{R}$)
- Imaginary Numbers

$\sqrt{-1}$

i

NUMBER THEORY

- Counting Numbers ($\mathbb{N}$)
- Whole Numbers ($\mathbb{W}$)
- Integers ($\mathbb{Z}$)
- Rational Numbers ($\mathbb{Q}$)
- Irrational Numbers
- Real Numbers ($\mathbb{R}$)
- Imaginary Numbers
- Complex Numbers

Real $\cup$ Imaginary

NUMBER THEORY

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- Integers ($\mathbb{Z}$)
- Rational Numbers ($\mathbb{Q}$)
- Irrational Numbers
- Real Numbers ($\mathbb{R}$)
- Imaginary Numbers
- Complex Numbers
- Odd & Even Numbers

NUMBER THEORY

- Counting Numbers ($\mathbb{N}$)
- Whole Numbers ($\mathbb{W}$)
- Integers ($\mathbb{Z}$)
- Rational Numbers ($\mathbb{Q}$)
- Irrational Numbers
- Real Numbers ($\mathbb{R}$)
- Imaginary Numbers
- Complex Numbers
- Odd & Even Numbers
- Prime Numbers

2, 3, 5, 7, 11, 13, …

NUMBER THEORY

- Counting Numbers ($\mathbb{N}$)
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- Real Numbers ($\mathbb{R}$)
- Imaginary Numbers
- Complex Numbers
- Odd & Even Numbers
- Prime Numbers
- Composite

NUMBER THEORY

- Counting Numbers ($\mathbb{N}$)
- Whole Numbers ($\mathbb{W}$)
- Integers ($\mathbb{Z}$)
- Rational Numbers ($\mathbb{Q}$)
- Irrational Numbers
- Real Numbers ($\mathbb{R}$)
- Imaginary Numbers
- Complex Numbers
- Odd & Even Numbers
- Prime Numbers
- Composite
- Perfect Number

6 (1, 2, 3),

28 (1, 2, 14, 4, 7),

496

TYPES OF NUMBERS

COUNTING NUMBERS

- are positive whose numbers excluding zero.

WHOLE NUMBERS

- are positive integers including zero.

INTEGERS

- are numbers formed by the natural numbers including zero together with the negatives of the non – zero natural numbers.

RATIONAL NUMBERS

- are numbers that can be written as fraction and whose numerators and denominators are integers provided that the denominator is not equal to 0.

IRRATIONAL NUMBERS

- are numbers that can be written as non terminating and non repeating decimal.

REAL NUMBER

- are numbers comprised of all rational and irrational numbers.

IMAGINARY NUMBERS

- is the square root of negative one. Any real number times *i* is an imaginary number.

COMPLEX NUMBERS

- are the combination of real numbers and imaginary numbers.

ODD NUMBER

- is a number when divided by 2 contains a remainder of 1.

EVEN NUMBERS

- is a number divisible by 2.

PRIME NUMBERS

- is a natural numbers greater than 1 that has no positive divisors other than1 and itself.

COMPOSITE NUMBERS

- is a positive integer which has a positive divisor other than 1 or itself. In other words a composite number is any positive integer greater than 1 that is not a prime number.

PERFECT NUMBERS

- is a positive integer that is equal to the sum of its proper positive divisors, that is, the sum of its positive divisors excluding the number itself.